

Triadic Modular Holography and Dimensional Mechanics

Documents and Contents for Academic Transfer

Oumar Haidara Fall · Rubén Ramos Balsa

Triadic Modular Holography and Dimensional Mechanics develop a unified explanation of the origin of constants and the relations underlying physics. The discrete construction of the continuum brings number, geometry, symmetry and physical magnitude into a common genealogy. It addresses a historical question in the relationship between mathematics and physics: what determines fundamental values, and what organisation they express. Explaining a constant encompasses its generation, the relations fixing its value and its role in transformations. Structure determines mathematical consequences; metrology tests their realisations in physical units.

The APP–TRIT–TPK kernel articulates this construction. APP brings addition and multiplication together on a common support; TRIT introduces orientation and a ternary regime; the Topological Phase Kernel transports configurations with memory, carry, boundary and path. In the nonadic connection, phase returns while memory advances. Each compatible extension increases resolution and preserves the dependencies of preceding levels. The generation of a number determines its value and retains the relations it transmits to subsequent constructions.

The central thesis is the double projection of a single discrete structure of the continuum. Refinement of oriented half-open cylinders reduces their diameters and determines real coordinates; compatibility between local domains reconstructs the Archimedean line. The conjugate projection preserves boundary incidences and their exceptional stabilisers. Measurement and symmetry express two related aspects of the same constitution: every value retains a geometric origin, an orientation and a memory. This unity connects arithmetic, geometry and physical organisation.

From this genealogy, π , ϕ and e are generated to arbitrary depth and α is derived through two internal routes. The electron connects algebraic structure, action and observable mass. Outward and return sections determine the Planck scale; the memory complement and α construct a length whose area, composed with action and propagation, determines $G = c^3 L^2 / \hbar$. Length precedes the gravitational coefficient. Planck, Boltzmann, permeability and permittivity are connected with Barbero–Immirzi and entanglement entropy through explicit operations and common dimensional realisations.

The holographic constant concentrates this continuity. Exceptional and temporal compatibility determines a unique record in the terminal domain. The same record enters the arithmetic closure of α , reversible information conservation and the incidences leading to Steiner, Golay and Leech. The chain extends to vertex operator algebras, Moonshine and the Monster. Modular reciprocity and dimensional decompositions articulate the connections with T-duality and M-theory, preserving the common origin of the representations.

The general mass law converts the persistence of configurations into positive operators, state classification and spectra. It brings particles, paths, orbits and mass readers into a common construction. The hexad–octad transition relates equal area to different sector composition and entropy, connecting information and quantum mixing. Bose–Einstein, Fermi–Dirac and the radiation laws extend the construction to collective behaviour. Gravitation, primes, the zeta function, Grassmannians and cardinality develop the argument further. Magnitudes express structure; understanding an entity means explaining its constitution, persistence and possibilities of transformation.

Organization and Purpose of the Material

This collection brings together the knowledge developed in Triadic Modular Holography and Dimensional Mechanics for study and academic transfer. Its texts, programs, data, tables and digital certificates make it possible to examine the operations, reproduce the calculations and follow dependencies from the foundations to their mathematical and physical consequences.

The integral corpus and the systematic treatise are the principal references. The former assembles foundations and developments produced with the assistance of artificial intelligence systems under the authors' direction; the latter sets out the operations and proof steps in greater detail. Two shorter, complementary routes facilitate access to this foundational material. The time-crystal monograph studies the double projection of the continuum and the relationship between periodicity and memory. The narrative exposition places the historical and conceptual problem in context and explains the significance of its mathematical and physical consequences.

The twelve monographs extract constructions, proofs, and applications from the foundational documents, following their dependencies. I generates π , ϕ , and e , derives α , and develops the algebraic structure of the electron. II constructs the Planck sections, angle and counter-angle, Barbero–Immirzi, entanglement, and CKM mixing; it articulates the thermal realization of Boltzmann and the connection between Planck length and gravitation. III derives Catalan's constant, permittivity, and permeability; it reconstructs the constitutive function and obtains impedance, propagation speed, and quantum electrical relations.

IV develops the arithmetic origin of exceptional symmetries from the difference between sum and product. It follows the incidences of the formal kernel through Witt, Golay, Leech, Moonshine and the Monster, and articulates their extensions towards T-duality and M-theory. V obtains the Bose–Einstein and Fermi–Dirac distributions and radiation laws. VI defines a particle through persistence, classifies states and paths, and constructs the general mass law and its operator; the annex brings together states, paths, masses and widths.

VII proves compatibility and restitution between action, length, gravitation, propagation, mass and information while preserving the order of transformations. VIII develops the gravitational and torsional response, field equations and the transition from contraction to expansion. IX connects primes, the zeta function and Weil positivity. X constructs positive Grassmannian and amplituhedral representations, lattice duality and the Yang–Mills mass gap, with their length and area readings. XI constructs the real line discretely and classifies its internal subsets within holographic closure. XII determines the holographic constant and the laws of information conservation, recovery and reversibility.

The collection makes an extensive investigation accessible in the service of mathematical and physical knowledge. It is offered for study, reproduction, assessment and academic use, and as a basis for possible scientific publications. The exposition connects operations, objects and relations with their physical realisations and metrological comparison; the proofs and executable materials allow the content, its provenance and its scope to be studied together.

Reference Documents of the Fundamental Corpus

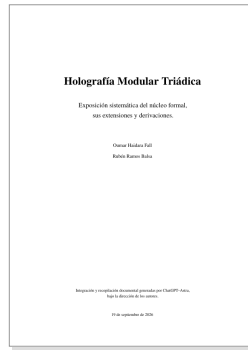
Triadic Modular Holography · Dimensional Mechanics



El cierre holográfico del infinito

La estructura discreta del continuo y el origen de las constantes fundamentales

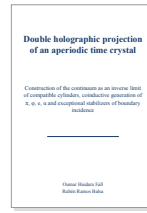
2.249 pages



Triadic Modular Holography

Systematic exposition of the formal kernel, its extensions and derivations.

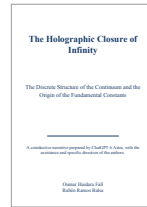
3.960 pages



Double holographic projection of an aperiodic time crystal

Construction of the continuum as an inverse limit of compatible cylinders, coinductive generation of π , ϕ , e , α and exceptional stabilizers of boundary incidence

858 pages



The Holographic Closure of Infinity

The Discrete Structure of the Continuum and the Origin of the Fundamental Constants · A coinductive narrative prepared by ChatGPT 6 Astra, with the assistance and specific direction of the authors.

327 pages

Series of Monographic Developments

Constructions, Proofs, and Applications Organized for Academic Study

I



Coinductive generation of π , ϕ and e to arbitrary depth, derivation of α and the algebraic structure of the electron

202 pages

II



Structural origin of the Barbero-Immirzi parameter and CKM mixing angles; derivation of the Planck, Boltzmann, and universal gravitational constants

Relationship between action, area, information, and entanglement entropy

286 pages

III



Constitutive structure of the vacuum and electromagnetic propagation

Derivation of Catalan's constant, permeability, and permittivity

208 pages

IV



Extension of the discrete structure of the continuum: vertex operator algebras, Moonshine, T-duality, and M-theory

218 pages

V



Quantum Statistics and Radiation

Structural Generation of the Bose-Einstein and Fermi-Dirac Distributions

209 pages

VI · Appendix



Particles, Persistence and Derivation of the Mass Spectrum

172 pages

Appendix · Structural and Metrological Catalogue of Particles

463 pages

VII



Structural Relations among Physical Constants

Action, Geometric Reciprocity, and Memory of Transformations

385 pages

VIII



Gravitation, torsion, and cosmological dynamics

Arithmetic memory, field equations, and transition from contraction to expansion

241 pages

IX



Genealogical arithmetic of prime numbers and spectral structure of the Riemann zeta function

Arithmetic transport, spectral moments, and Weil's positivity criterion

193 pages

X



Positive Grassmannians and amplituhedra: canonical forms and refinement

Lattice duality and the Yang-Mills mass gap in Triadic Modular Holography

278 pages

XI



Resolution of the Continuum Hypothesis in the Holographic Closure

Discrete Construction of the Real Line $\mathbb{R}$ and Cardinal Classification of Internal Subsets

200 pages

XII



Holographic Extreme and Mean Ratio

The holographic arithmetic-geometric coupling constant. Conservation and reversibility of information

540 pages

Reference Documents of the Fundamental Corpus



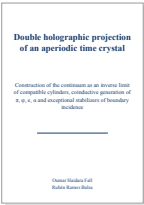
El cierre holográfico del infinito
La estructura discreta del continuo y el origen de las constantes fundamentales
2.249 pages

The comprehensive corpus develops the foundations of Triadic Modular Holography and Dimensional Mechanics: the discrete constitution of the continuum links number, geometry, and physical quantity. Starting from APP–TRIT–TPK and the joint construction of the numerical and exceptional projections, it develops the constants, the structure of the electron, the mass operators, and the realizations of action, gravity, and information. Definitions, theorems, examples, and verification procedures allow the dependencies between results to be reconstructed. It preserves the elementary constructions and their specialized developments, including cardinal closure within holographic closure. It is the technical reference for studying how a common genealogy transmits structure between results, within the domain specified for each claim.



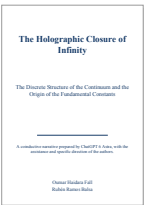
Triadic Modular Holography
Systematic exposition of the formal kernel, its extensions and derivations.
3.960 pages

The treatise systematically sets out the operations of the formal core, its extensions, and its derivations. It starts from the initial grid and develops compatible prolongation, double projection, and the correlated generation of constants. The selection of the coupling record, its recovery, exceptional incidence, and the laws of information conservation receive explicit treatment. Dimensional Mechanics extends these foundations towards action, area, physical constants, and material organization. The exposition shows what each transformation determines, what it preserves, and what it allows to be reconstructed; the proofs specify their conditions. The intermediate steps, developed in detail, make the transmission of structure from elementary operations to subsequent realizations intelligible.



Double holographic projection of an aperiodic time crystal
Construction of the continuum as an inverse limit of compatible cylinders, coinductive generation of π , ϕ , e , a and exceptional stabilizers of boundary incidence
858 pages

The monograph develops the double projection of the continuum from an arithmetic of oriented configurations. Refinements preserve the preceding levels while increasing resolution, height, and memory; their compatible sequences form an inverse limit. The aperiodic time crystal expresses the coexistence of local phase returns with an ongoing global evolution. On this support, π , ϕ , e , and a are generated, the electronic section is constructed, and action, area, and information are linked. The conjugate projection preserves the incidences that lead to exceptional realizations. Numerical determination and temporal order are thus linked to the same structure that preserves the geometric relations of the continuum. The extension of the dodecaphasic profile establishes uniform selection of the first superstable centres and their parametric Feigenbaum limit, preserving branch orientation and folding memory.



The Holographic Closure of Infinity
The Discrete Structure of the Continuum and the Origin of the Fundamental Constants · A coinductive narrative prepared by ChatGPT 6 Astra, with the assistance and specific direction of the authors.
327 pages

The narrative exposition develops the meaning of a construction that links mathematical explanation, measurement, and physical constitution. It asks what determines a constant, what distinguishes a particle, and what relations an organization preserves as it transforms. The historical and philosophical background leads to APP–TRIT–TPK, the continuum, the constants, the electron, and the organization of masses. Exceptional incidence, area, and information reveal connections across these topics. The narrative places each result within the common argument and explains what knowledge it contributes: how generative relations determine properties and make them intelligible. This epistemological and ontological approach offers complementary access to the technical documents containing the corresponding constructions and proofs.

Series of Monographic Developments



I. Coinductive generation of π , ϕ and e to arbitrary depth, derivation of α and the algebraic structure of the electron

202 pages

Angular, modal, and propagation readings of APP–TRIT–TPK generate π , ϕ , and e by compatible prolongation to arbitrary depth. Dodecaphase closure preserves remainders, orientation, and carry and determines α through two internal routes. Exceptional–temporal selection fixes a unique record; cubic completion, $729 + 271 = 1000$, and its incidence explain its role. Five regions retain the same tritic word without losing their oriented distinctions. For the electron, projector incompatibility gives $\sqrt{3}/4$; displacement and autocorrelation readers agree on (80, 54, 6), while torsional and return composition determines the mass evaluation. Integer inversion of refinement recovers previous levels: number, incidence, and memory remain linked before subsequent dimensional realizations. Comparing three regional Witt stars specifies incidence polarisation and the rigidity of the frame joining the supports of α , e , ϕ and APP.



II. Structural origin of the Barbero–Immirzi parameter and CKM mixing angles; derivation of the Planck, Boltzmann, and universal gravitational constants

Relationship between action, area, information, and entanglement entropy

286 pages

The constructed action distinguishes outward, return, and common Planck scales. Angle and counter-angle articulate the hexad–octad transition: its 90/120 incidences determine the Barbero–Immirzi parameter and the invertible rational CKM reader. The three mixing angles, CP phase, and Jarlskog invariant are obtained and explicitly compared with PDG 2025. The energy of the 108-step cycle links action and tritic information. The marked thermal realization preserves sectors of dimensions 1, 380, and 649, their grading, and memory; its spectral intensity determines the coefficient 1.380649×10^{-23} and the corresponding transducer through additive information and conditioned energy. The length $L\alpha = (5759/23040)\alpha^1 L^*$ precedes G : conjugate readings fix $G = c^3 L\alpha^2/h$ and preserve $\hbar G/c^3 = L\alpha^2$. Action, area, mixing, and temperature express distinct operations of the same construction.



III. Constitutive structure of the vacuum and electromagnetic propagation

Derivation of Catalan's constant, permeability, and permittivity

208 pages

The electromagnetic response is constructed through two oriented channels of the same transport. The incidences 90 and 120 determine an operator whose eigenvalues produce the constitutive coordinates of permittivity and permeability; impedance and propagation speed are obtained from them. Cubic inversion recovers the initial channels. The double moment of the resolvent associated with the transport generates Catalan's constant; the Pell specialization establishes an exact identity between that moment and two reciprocal factors. The dimensional realization then allows charge, von Klitzing resistance, conductance, magnetic flux, and the Josephson constant to be derived. Their common identities make explicit the relations preserved in passing from the internal construction to measurable quantities. The extension makes inverse restitution from the constitutive responses explicit: regularity at the origin fixes the oriented Catalan function while preserving its arguments and transport channels.



IV. Extension of the discrete structure of the continuum: vertex operator algebras, Moonshine, T-duality, and M-theory

218 pages

The prolongation of the core towards exceptional realizations preserves marked incidence and the memory of routes. The complete census distinguishes oriented conditions, emissions, and ternary images, preserving their multiplicities. A Witt flag determines a self-dual ternary code; its gluing leads to a Niemeier lattice and to an even, unimodular, rootless neighbour identified as the Leech lattice. On this lattice, the vertex operator construction reaches the Moonshine module, whose graded character is $j - 744$ and whose automorphism group is the Monster. The nonadic tower yields the operator continuum. T-duality, dimensional screens, and the dynamical realization of M-theory are articulated through maps and domains that preserve the provenance of the initial record.

Series of Monographic Developments



V. Quantum Statistics and Radiation

Structural Generation of the Bose–Einstein and Fermi–Dirac Distributions

209 pages

The compositional parity of states with trajectories and memory determines the symmetric and exterior occupation completions. The latter establishes algebraic Pauli exclusion; the factorization of grand-canonical equilibrium produces the Bose–Einstein and Fermi–Dirac distributions. In the bosonic electromagnetic realization, occupations lead to the Planck spectrum, the Stefan–Boltzmann flux, and the Wien maxima. The Hamiltonian of the unitary evolution reproduces the calendar shift, with a 108-state Floquet realization. Complementary observables, correlators, and a stationary maximum-entropy measure specify information: the areal–volumetric partition reaches $\log \phi$, while a relative-divergence identity quantifies the deficit of other distributions. Composition with length and action preserves the areal scale. Area and the modular Hamiltonian recover occupations through an invertible system with determinant 30.



VI. Particles, Persistence and Derivation of the Mass Spectrum

Structural definition of a particle, classification of states, and construction of the mass operator

172 pages

Persistence over compatible prolongations defines the particle, preserving orientation, sheet, spin, flavour, colour, and memory. The classification organizes 81 cells, 56 sets of routes, 13 multisections, and 324 oriented conditions, with a distinguished central electronic section. On each fibre, a congruence transformation constructs the positive mass operator. In the electron, the displacement and autocorrelation readers coincide at (80, 54, 6) and produce an explicit scalar evaluation. Composition acts on complete records, with links and boundary conditions, before extracting the signature. The realizations of conjugation, occupation, and resonance specify what remains when two states share a mass reading and what information distinguishes their structure and dynamics. The revised longitudinal scale preserves this law and its positivity: changing the action section transports the mass form and reader jointly, retaining physical evaluations.

Documentary Appendix. Structural and Metrological Catalogue of Particles. States, Routes, Masses, and Widths. 463 pages

The catalogue makes the structural and metrological domain accompanying the study of particles available for consultation. It brings together the 324 oriented routes in entries that link the conditions of construction, the readers, and the corresponding quantities. It distinguishes phase return, cell, orbit, and kinematic state and organizes 13 multisections, 56 sets of routes, and 23 inventory classes. The documentation comprises 471 masses and 384 widths, with units, sources, and observational status. It allows each correspondence between route, quantity, and reference to be examined while preserving the differences between central values, intervals, limits, and widths. These counts describe the contents of the catalogue, not the number of individual predictions.



VII. Structural Relations among Physical Constants

Action, Geometric Reciprocity, and Memory of Transformations

385 pages

This monograph proves compatibility and restitution relations between action, propagation, mass, and information. Speed and impedance recover the spectrum of the constitutive channels; the Barbero–Immirzi functional distinguishes their oriented distribution, and the Catalan moment preserves the functional response. Composing elliptic transformations restores the action coefficient and records the order of operations. The marked length and its memory complement connect action and gravitation through a conserved area. In the thermal realization, the cycle supplies energy and the spectral reference supplies the informational weight; both are transported with the area and horizon section. A mass coordinate relates frequency and reciprocal lengths. These identities explain how constants are reconstructed from one another while preserving orientation, units, and the domain of each operation. The total action joins geometry, the strong and electroweak sectors and chiral matter, with compatible backreaction and constraints in the regular realisation. Horizon relations link area, entropy and temperature and preserve their joint transformation with action.



VIII. Gravitation, torsion, and cosmological dynamics

Arithmetic memory, field equations, and transition from contraction to expansion

241 pages

Transported memory is realized geometrically through Cartan routes, a variational screen, and a matter current. Eliminating contorsion determines the explicit torsional amplitude of the spinorial sector; an occupation sector retains its positivity even with a vanishing mean current. Spatial growth and memory produce an a^{-6} dilution, which enters the transition from contraction to expansion with a unique, transverse threshold under the stated homogeneous conditions. For spatially flat dust, the exact solution is regular at all times and causally complete. The five-component carrier is linked to the tensorial and transverse readings; horizon balances, reduced information, and advanced and retarded propagation develop the response through explicit operators and boundary conditions. The Planck length is constructed from α and the memory complement; composition with action and propagation determines the gravitational coefficient. Accumulated memory is controlled by its total variation, preserving orientations that a net balance would cancel.

Series of Monographic Developments



IX. Genealogical arithmetic of prime numbers and spectral structure of the Riemann zeta function

Arithmetic transport, spectral moments, and Weil's positivity criterion

193 pages

The native product organizes normal-form words and irreducibles whose evaluation identifies the positive primes. Their occupations and traces construct the Euler product and the prime-power moments; the theta–Mellin transform produces the completed function with gamma factors, polar terms, and a functional equation. The boundary reading preserves quotients, vacancies, and memory; the double circle distinguishes phase compression from recoverable information. For the Weil form, the exact elimination of details through the Schur procedure preserves the infinite complement and expresses positivity through an equivalent matrix condition. In the complete nonadic window, a uniform coercive bound is proved, preserved under translation and interior refinement. Arithmetic genealogy, reader reconstruction, and spectral control of infinite functional domains are thus connected.



X. Positive Grassmannians and amplituhedra: canonical forms and refinement

Lattice duality and the Yang–Mills mass gap in Triadic Modular Holography

278 pages

The generated record is transformed into a positive Grassmannian representation with a marked inverse that allows value, incidence, and readers to be recovered. Nonadic refinement constructs flags and amplituhedral realizations at finite rank and finite external dimension; in rank one, it proves simplicial coverage and cancellation of interior residues. The moments of the same chronological measure produce Jacobi operators, simple spectra, and continued-fraction resolvents. The exceptional deformation preserves covolume and ternary symmetry, linking parameter inversion, lattice duality, and theta reciprocity. The gauge development connects projective measure, action, reflection positivity, coercivity, and Hamiltonian reconstruction, with a persistent witness of the spectral gap. The hexad–octad transition provides complementary area and occupation readers and a finite entanglement identity. The longitudinal scale induces a common positive areal scale: it normalises the physical readers without changing Plücker signs, canonical residues or marked refinement.



XI. Resolution of the Continuum Hypothesis in the Holographic Closure

Discrete Construction of the Real Line $\mathbb{R}$ and Cardinal Classification of Internal Subsets

200 pages

Compatible routes construct the Archimedean line; the second projection preserves incidence, orientation, and memory. Exhaustive cylinder coverage links this line with the internal power set of the natural numbers. Within complete genealogical closure, condensation assigns each value an injective ordinal code; the coding of countable well-orders and their ternary lifting construct the injection in the opposite direction. Cantor–Bernstein establishes cardinal equality, and the cofinality argument classifies all internal subsets as countable or equinumerous with the line. The local branching into 729 subcylinders extends to arbitrary depth: phase returns and memory advances. The coupling record, correlated constants, and exceptional realization preserve the shared provenance of this construction.



XII. Holographic Extreme and Mean Ratio

The holographic arithmetic–geometric coupling constant.

Conservation and reversibility of information

540 pages

Exceptional incidence and the chronology of the readings jointly determine the coupling constant: the terminal domain moves from 19,446 candidates to 79 and finally to a unique record. The Hadamard transform provides its reversible representation; the positional reading enters the closure of α , and the incidence reading prolongs the construction towards Leech and Moonshine. A bilateral law for two isometric transports establishes conservation in arbitrary dimension. Refinement with carry admits exact inversion; the archive preserves the norm and reconstructs the state to unlimited depth. The geometric realizations transport the direction of the record through determinant-one deformations, preserving Plücker signs and covolume. Selection, conservation, and restoration thus link observed information and memory. The retained complement connects information conservation, marked length and reciprocity between action and gravitation without changing the exceptional–temporal selection of the record.

Constants, Fundamental Relations, and Physical Data

Joint Construction of the Continuum and the Constants

Generation begins in the arithmetic nucleus and preserves the sequence of its transformations. The coordinates π , φ , e , and α belong to the same genealogy; conventional values are not introduced as generating data. Sources: article I and the time-crystal monograph.

Arithmetic Nucleus, Orientation, and Memory

APP compares addition and multiplication over the alphabet $1, \dots, 9$ in two orientations; TRIT preserves the three local regimes, and TPK incorporates phase, memory, and carry. The census traverses 104,976 ordered pairs of oriented cursors (324^2), distinguishes 468 emissions, and obtains 243 ternary images within an ambient set of 729 elements. Each number counts objects at a different level.

Compatible Prolongation and the Nonadic Connection

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

The L_0 lifts lead to w_{12} and w_{18} ; the L_1 lifts change regime at w_{24} and w_{30} while preserving prefix, orientation, and memory. R_{36} opens the coinductive boundary, and G_9 brings together nine local domains. The holonomy Γ_9 represents a complete turn: phase returns and the memory counter advances. The monodromic representation preserves this displacement between turns; even when the 108-position calendar is restored, accumulated memory remains. Nested cylinders determine the coordinates through compatible prolongation.

Sturmian Order and Vacancies

$$\delta = \frac{\log(10/9)}{\log 10}, \quad z_n = \lfloor (n+1)\delta \rfloor - \lfloor n\delta \rfloor$$

$$C_n - C_{n-1} = 1 - z_n$$

The comparison of ternary and decimal capacities produces an aperiodic Sturmian reading. Its events are separated by 21 or 22 steps; the distribution of short intervals induces a second scale of 6 or 7. A vacancy adds a block to the record without increasing decimal capacity. The balance preserves acquired capacity and recorded vacancies, linking refinement and the tritic regime. This capacity reading does not replace the full dynamics of TPK.

π , φ , and e · Three Readings of the Same Process

Coinductive generation determines π , φ , and e by prolonging the same process to arbitrary depth: every refinement preserves what has been constructed and adds resolution. For π , one transverse, three positive, and one negative region converge to the ternary sequence 010211, with multiplicities $432 + 4 \times 144 = 1,008$. The marked flag links these regions to four residual octads and one transverse octad of the Witt star. Modal incidence determines φ and compatible propagation determines e . The finite verification covers 396 levels, 2,376 trits, 43 turns, and 387 equal-phase pairs per channel, retaining orientation, carry, and memory together. Analytic characterizations subsequently recognize the generated constants.

$\pi = 3.141592653589793238462643383279\dots$; $\varphi = 1.618033988749894848204586834365\dots$; $e = 2.718281828459045235360287471352\dots$

Imaginary Unit and Refinement

$$J_E^2 = -I, \quad N_{n+1} = 729N_n + \nu(b_{n+1})$$

The negative-square operator realizes complex orientation. The ternary update increases resolution without erasing the preceding levels.

α · Closure with Carry and Analytic Reading

$$\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}$$

$$E_\infty(X) = E_9(X) + \lambda \frac{X^{10}}{1 - X^{3/729}}, \quad E_\infty(\alpha_B) = 0$$

Exceptional–temporal selection fixes the coupling record. The first route normalizes remainders and carries; the second completes the vacancy reader with the same pre-coordinate. Compatibility establishes $\alpha_A = \alpha_B$. The coefficient λ is constructed from that pre-coordinate: these are not two independent fits.

$\alpha = 0.007297352569283800997285105472\dots$

Constants, Fundamental Relations, and Physical Data

Action and the Algebraic Structure of the Electron

Action and the electron evaluation are presented with their complete factors, following the dependency order of the corpus. Sources: articles I, II, and VI.

Reduced Planck Constant · Action before Return

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5$$

$$\hbar_{\text{pre}} = H_5 \cdot 10^{-34} \mathcal{U}_S, \quad h_{\text{pre}} = 2\pi\hbar_{\text{pre}}$$

H_5 combines the contributions of α and φ in the action reading before return. Its dimensional realization first gives reduced action $\hbar_{\text{pre}}$; a complete turn supplies the factor 2π and produces h_{pre} . The subscript pre identifies this orientation of the trajectory.

For $US = 1 \text{ J s}$: $\hbar_{\text{pre}} = 1.054571817646154469... \times 10^{-34} \text{ J s}$; $h_{\text{pre}} = 6.626070149999987926... \times 10^{-34} \text{ J s}$.

Regional Return and Action Ratio

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} C_\pi(\alpha), \quad R_{\text{act}} = H_5 / \eta_{\text{ret}}$$

$$C_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1-D_A\alpha}, \quad D_A = \exp(-100\pi\alpha/9)$$

Five emissions retain the terminal block (169, 272, 272) four times and its alternative once. Within the persistent block, the coordinate distinguished under its preserving permutations selects 169. Word length supplies degree six, and the transport determinant generates the regional continuation. This determines C_π and hence the returned action η_{ret} . Ract compares outward and return readings: its excess over one, approximately 6.93×10^{-10} , measures an internal asymmetry that enters the mass law through state-specific exponents.

$R_{\text{act}} = h_{\text{pre}}/h_{\text{ret}} = 1.0000000006932817630...$

Reduced Planck Constant · Returned Action

$$\hbar_{\text{ret}} = \eta_{\text{ret}} \cdot 10^{-34} \mathcal{U}_S, \quad h_{\text{ret}} = 2\pi\hbar_{\text{ret}}$$

The subscript ret denotes the returning trajectory. This section transports the regional contribution that distinguishes the two orientations and is used in the gravitational evaluation shown here.

$\hbar_{\text{ret}} = 1.054571816915039061... \times 10^{-34} \text{ J s}$.

Planck · Common Scale of Outward and Return Action

$$\hbar_{\text{geom}} = \sqrt{\hbar_{\text{pre}}\hbar_{\text{ret}}}, \quad h_{\text{geom}} = 2\pi\hbar_{\text{geom}}$$

$$\hbar_{\text{pre}} = \hbar_{\text{geom}} e^{\xi}, \quad \hbar_{\text{ret}} = \hbar_{\text{geom}} e^{-\xi}, \quad \xi = \frac{1}{2} \log R_{\text{act}}$$

The geometric mean separates the common action scale from the orientation of return. It is the parametrization developed in the integral corpus, equation (36.14); the two orientations remain explicit. Reduced action is given first, and multiplication by 2π then gives the action of a complete turn.

$\hbar_{\text{geom}} = 1.054571817280596765... \times 10^{-34} \text{ J s}$; $h_{\text{geom}} = 6.626070147703121129... \times 10^{-34} \text{ J s}$.

Electron · Complete Evaluation Expression

$$\varepsilon_e^\beta = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4) R_{\text{act}}^{80-54\alpha+6\Delta_4}$$

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right), \quad m_e^\beta = \varepsilon_e^\beta u_E / c^2$$

Projector incompatibility contributes $\sqrt{3}/4$; torsional normalization and the return incorporate the profile (80, 54, 6). With $u_E = 1 \text{ MeV}$, the coordinate is realized as energy and mass. The complete numerical comparison appears in the metrological table.

$m_e\beta = 0.510998950690000808608... \text{ MeV}/c^2$.

Constants, Fundamental Relations, and Physical Data

Action and the Algebraic Structure of the Electron · continued

Mass Operator and Organization of States

$$M_M^{\beta, r} = R_{\text{act}}^{B_M^r/2} M_M^{(0)} R_{\text{act}}^{B_M^r/2} \geq 0$$

Mass is obtained from the persistent organization of the state: route, orientation, occupation and memory determine its signature. $M^{(0)}$ is the initial mass form; B records the response to return and R_{act} transmits the action ratio already constructed. Applying the same factor on both sides preserves a positive mass operator. Its evaluation on each state produces the corresponding scalar reading. The 81 cells, 56 route sets, 13 multisections and 324 oriented conditions organize this domain; links and boundaries are preserved before the signature is extracted.

Length, Planck Scales, and Gravitational Coupling

The marked length is constructed before G . Action and the conjugate readings of the same positive character determine its coupling. Speed retains its constitutive provenance $c = \hat{c} \text{ uC}$; dimensional bases are restored only once.

Planck Length and Gravitational Coupling

$$\begin{aligned} N &= 12 \cdot 4 \cdot 120 = 5760, & L_\alpha &= \frac{N-1}{4N} \alpha^{16} L_* \\ H\Lambda &= \hbar_a c I, & R\Lambda &= L_\alpha^2 I, & M &= H/c^2, & R &= GM/c^2 \\ G_a &= \frac{c^3 L_\alpha^2}{\hbar_a}, & \frac{G_{\text{pre}}}{G_{\text{ret}}} &= R_{\text{act}}^{-1}, & \frac{\hbar_a G_a}{c^3} &= L_\alpha^2 \end{aligned}$$

The observed return contributes $(N-1)/(4N)$; the complement $1/(4N)$ remains in memory. Longitudinal composition with α^{16} fixes L_α before G is evaluated. Energy and radial response preserve the same character: the two product identities fix their proportionality. The reduced gravitational-radius reading $R = GM/c^2$ then determines a unique G in this realization. With $L_* = 1 \text{ m}$, $\hbar_{\text{ret}} = \eta_{\text{ret}} \times 10^{-34} \text{ J s}$, and $c = 299,792,458 \text{ m/s}$, one obtains the displayed values. The circular expression is recovered with $t_0 = \pi L_\alpha / (54c)$; the factor $54/\pi$ is not applied to L_α a second time.

$L_\alpha = 1.616254982625126330... \times 10^{-35} \text{ m}$; $G_{\text{ret}} = 6.674299659414022706... \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.

Radial Readings and Scales of the Same Character

$$\begin{aligned} Y > 0, & \quad R = L_\alpha Y, \quad \Lambda = L_\alpha Y^{-1}, \quad H = \frac{\hbar_a c}{L_\alpha} Y \\ t_P &= \frac{L_\alpha}{c}, \quad m_{P,a} = \frac{\hbar_a}{c L_\alpha}, \quad E_{P,a} = \frac{\hbar_a c}{L_\alpha} \\ t_0 &= \frac{\pi}{54} t_P, \quad 108 t_0 = 2\pi t_P \end{aligned}$$

On a positive, invertible finite-dimensional fibre, the readings preserve $H\Lambda = \hbar_a c I$ and $R\Lambda = L_\alpha^2 I$. For $y > 0$, they give $m = m_P y$, $rg = L_\alpha y$, and $\tilde{\lambda} = L_\alpha / y$. Equality of the two lengths characterizes $y = 1$, the circular specialization of the construction. The Schwarzschild radius is $2rg$. Changing the action section by a factor ρ multiplies m_P and E_P by ρ , divides G by ρ , and preserves L_α and t_P . The null sector retains its separate treatment.

Boltzmann Constant · From Tritic Decision to Energy

$$\begin{aligned} I_{\text{TRIT}} &= - \sum_{j=1}^3 \frac{1}{3} \log \frac{1}{3} = \log 3, & T_{108} &= 108 t_0 \\ \varepsilon_a &= \hbar_a \frac{2\pi}{T_{108}} = \frac{\hbar_a}{108 t_0} = \frac{\hbar_a c}{L_\alpha} \end{aligned}$$

The uniform ternary decision has three equiprobable alternatives, whose information is $\log 3$. The 108-step cycle supplies duration; the constructed action converts its frequency into energy. Thus $\varepsilon_a / \log 3$ is energy per unit of information. The index a preserves outward or return orientation. Information, action, length, and propagation are connected before temperature is expressed in kelvin.

Constants, Fundamental Relations, and Physical Data

Length, Planck Scales, and Gravitational Coupling · continued

Boltzmann · Spectral Reference and Thermometric Selection

$$b^* = 10^{-23} \left(1 + \frac{380}{1000} + \frac{649}{1000^2} \right) = 1.380649 \times 10^{-23}$$

$$S_B = b^* s(\rho), \quad \mathcal{E}_A = \text{Tr}(\rho H), \quad \Theta_{B,A}^{-1} = \frac{dS_B}{d\mathcal{E}_A} = b^* \beta$$

$$b^* \Theta_{\text{clk}, a} = \frac{h_a}{108 t_0 \log 3}, \quad G_a b^* \Theta_{\text{clk}, a} \log 3 = c^4 L_a$$

Comparing 729 tritic words with 1000 decimal positions fixes a capacity: after the first vacancy, the marked return has $C_{24} = 23$. Twenty labels produce $20 \times 19 = 380$ off-diagonal relations; twenty-five positions and twenty-four adjacencies produce $25^2 + 24 = 649$ records. The realization assigns these sectors, together with the rank-one vacuum, to degrees 23, 26, and 29 of a reference with ratio 1/10. Their relative weight gives b^* . The instrument retains the complement and permits recovery of the state. Information per reference and energy conditioned on the sector form the thermal pair: additivity fixes the former, and for a finite nonscalar Hamiltonian their differential ratio determines temperature and transducer. Cut, grading, and the pair of readings specify the realization. The cycle supplies $\log 3$ while preserving action, area, and gravitation.

Coefficient of the marked realization: $b^* = 1.380649 \times 10^{-23}$.

Area, Information, and Quantum Mixing

The transition from hexad to octad preserves incidence relations and distinguishes sectorial composition and entropy. Sources: monographs II and X. The experimental CKM comparisons appear in the final metrological tables.

Barbero–Immirzi and the 90/120 Incidences

$$\gamma = \frac{\pi A C^*}{180} + 12 \Delta S_{90} - \Delta S_{120}$$

A and C^* are angular coordinates expressed in degrees; $\Delta S_n = S_n(q^-) - S_n(q^+)$. The sectors of 90 and 120 configurations arise from the hexad and octad incidences.

$\gamma = 0.274007672694459274747255\dots$; Ghosh–Mitra theoretical reference: $0.2740668582328300\dots$

CKM Angles and the CP Phase

$$\theta_{12} = 2A - 5h_\theta + 28\Delta, \quad \theta_{23} = 7h_\theta - \frac{25}{2}\Delta$$

$$\theta_{13} = A - 20h_\theta - \frac{2}{3}\Delta, \quad \delta_{\text{CP}} = 9A, \quad h_\theta = C^*/16$$

A, C^* , and Δ are expressed in degrees, with $\Delta = (180/\pi)\Delta_4$. The rational matrix relating the coordinates has determinant $-3857/6$: it allows the initial coordinates to be reconstructed from the mixing angles within the stated representation.

$\theta_{12} = 13.003313589509^\circ$; $\theta_{23} = 2.398056157332^\circ$; $\theta_{13} = 0.213977382244^\circ$; $\delta_{\text{CP}} = 65.676173123554^\circ$.

Meaning of Mixing and Angular Agreement

The three angles determine mixing between generations; the CP phase and Jarlskog invariant express its complex asymmetry. All five evaluations lie within the PDG 2025 reference intervals used here. In particular, the calculated phase 65.676173° is compared with $65.718259^\circ \pm 1.489690^\circ$. The final tables give both the sines and angles in degrees. Electroweak W/Z mixing is a distinct reading, displayed separately.

Area and Entanglement Entropy

The construction retains area, sectorial distribution, and boundary information together. These data distinguish equal-area configurations and articulate the readings of Barbero–Immirzi, entanglement, and mixing. Area measures geometrical extent; entropy records the distribution of the reduced state. Ghosh–Mitra supplies a second theoretical state-counting construction with which to compare γ .

Constants, Fundamental Relations, and Physical Data

Transport, Electromagnetic Constitution, and Statistics

The constitutive quantities arise from oriented transport; the occupation distributions arise from state completions. Sources: articles II, III, and V.

Catalan and the Reciprocal Pell Section

$$C_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad G_{\text{Cat}} = C_2(1)$$

$$C_2(2 - \sqrt{3}) = \frac{2}{3} G_{\text{Cat}} - \frac{\pi}{12} \log(2 + \sqrt{3})$$

The double moment of the resolvent generates Catalan's constant; the Pell section preserves two reciprocal factors of the same transport. Its full function enters the electromagnetic constitution: with $D = s \, d/ds$, $D^2 C_2(s) = s/(1+s^2)$ and $f(s) = (1-s^3)/(1-s^4) = (1+s+s^2)D^2 C_2(s)/[s(1+s)]$. The two channels are $r_{\pm} = f(q_{\pm}^{30})$. The inverse equation and the limit $C_2(0) = 0$ recover the oriented function, not only its extremal value.

$G_{\text{Cat}} = 0.915965594177219015054604\dots$

Permittivity, Permeability, Impedance, and Speed

$$r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}, \quad \hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = \frac{r_-}{r_+}, \quad \hat{c} = \frac{1}{r_+ r_-}$$

$$\mu = r_-^2 \frac{u_s}{u_c u_Q^2}, \quad \varepsilon = r_+^2 \frac{u_Q^2}{u_s u_c}, \quad Z = \hat{Z} \frac{u_s}{u_Q^2}, \quad c = \hat{c} u_c$$

The 90/120 incidences fix the eigenvalues of the two channels. All four internal coefficients are calculated. To express them in farads per metre, newtons per ampere squared, ohms, and metres per second, the formulas use the units of action uS , charge uQ , and speed u_c . The integral corpus also evaluates μ_0 , ε_0 , and Z_0 in SI units from α_{HMT} and the defining coordinates of action, charge, and speed; their values and the CODATA references are presented together in the numerical tables. The relations $\mu\varepsilon = c^{-2}$ and $\mu/\varepsilon = Z^2$ are preserved.

$\varepsilon = 0.999999257096636703\dots$; $\mu = 0.999448350479326935\dots$; $Z = 0.999724508538937215\dots$; $\hat{c} = 1.000276310486157526\dots$

Charge and Electrical Quanta

$$q_a = \sqrt{\frac{2ah_a}{Z}}, \quad R_K = \frac{Z}{2\alpha}, \quad G_0 = \frac{4\alpha}{Z}, \quad \Phi_{0,a} = \sqrt{\frac{h_a Z}{8\alpha}}, \quad K_{J,a} = \frac{2q_a}{h_a}$$

The construction links charge, resistance, conductance, flux, and the Josephson constant, preserving $RK \, G_0 = 2$ and $KJ \, \Phi_0 = 1$.

Bose–Einstein, Fermi–Dirac, and Radiation

$$\langle n_k \rangle_{\text{BE}} = \frac{g_k}{e^{\beta(E_k - \mu)} - 1}, \quad \langle n_k \rangle_{\text{FD}} = \frac{g_k}{e^{\beta(E_k - \mu)} + 1}$$

The symmetric and exterior completions generate bosonic and fermionic occupations. Their electromagnetic realization leads to the Planck spectrum and the Stefan–Boltzmann and Wien relations.

Relations between Constants and Restitution of Transformations

Monograph VII connects the readings of action, propagation, and mass through their common operations. The inverse relations recover outputs already constructed and preserve orientation, domain, and dimensional reference. Relations between constants extend a common construction towards the joint dynamics of geometry, fields and matter. Action–gravitation reciprocity preserves length, while thermal transport links energy and information.

Constitutive Compatibility and Reconstruction Sensitivity

In the normalized constitutive domain, speed fixes the product of the responses, and the Barbero–Immirzi functional distinguishes their oriented partition. Together, these data determine the labelled eigenvalues of the two channels. The logarithms of speed and impedance are derivatives of the same spectral potential; its Hessian characterizes the sensitivity of the inversion. Catalan's function and the constitutive function are recovered through a second-order differential equation with a condition at the origin. The spectral moment, arguments, and orientation are preserved jointly; the isolated decimal value of Catalan's constant does not replace them.

Constants, Fundamental Relations, and Physical Data

Relations between Constants and Restitution of Transformations · continued

Relative Transport and Restitution of Action

$$\rho_{\text{lazo}} = \left(\frac{501\alpha + \eta_{\text{ret}}}{499\alpha - \eta_{\text{ret}}} \right)^2, \quad 0 < \eta_{\text{ret}} < 499\alpha$$

Quarter-turns of conjugate elliptic forms produce a unimodular loop with reciprocal dilations. The stated branch and orientation allow the dimensionless action coefficient to be recovered. Closing the phase balance does not erase the imprint of the order of operations. Restoring a dimensional action additionally requires its scale to be preserved; conservation of symplectic area is not equivalent to an energy gain.

Mass, Frequency, Geometric Reciprocity, and Thermal Reading

$$\Xi_\Gamma = \frac{\Omega_\Gamma}{\omega_0} = \frac{r_{g,\Gamma}}{R_{108}} = \frac{R_{108}}{\lambda_\Gamma} = \frac{E_\Gamma}{k_B \Theta_{108} \log 3}$$

Under the specified circular and thermal conditions, a positive mass coordinate determines four readings. The two lengths dilate reciprocally and preserve the product R_{108}^2 ; the thermal cycle contributes the weighting $3^{(-\Xi_\Gamma)}$. The identity retains the mass constructor, its normalizations, and the convergence conditions: it does not identify different physical realizations without them.

Joint action and horizon reciprocity

$$S_A = \frac{k_B A}{4L_a^2}, \quad T_A = \frac{\hbar_a c}{2k_B \sqrt{\pi A}}, \quad S_M T_M = \frac{Mc^2}{2}$$

The same total action incorporates geometry, gauge fields and matter, with their sources and constraints in the regular domain. On a stationary Schwarzschild horizon, the constructed length determines the elementary area; entropy and temperature transform their action-section dependences reciprocally. Their product preserves the stated relation. Horizon and cycle temperatures belong to different realisations, related through their operators and conditions.

Gravitation, Memory, and Cosmological Dynamics

The coupling constructed from the marked length enters the gravitational response. Action reciprocity preserves the area, and material memory contributes the explicit torsional amplitude. Sources: monographs II and VIII and the joint development of length, memory, and gravitation.

Memory and Torsional Amplitude

$$s_0[Q, V_c] = \frac{3\kappa c^2 \hbar^2}{16} \frac{\gamma^2}{1 + \gamma^2} \frac{\text{Tr}(\varrho Q_{\text{sp}})}{V_c^2}, \quad \kappa = \frac{8\pi G}{c^4}$$

$$M_n = \varphi^{-2n} = (a_n/a_{\text{ref}})^{-6}$$

Eliminating contorsion produces the matter amplitude. In the occupation-three sector, $Q_{\text{sp}} = 4I$ preserves positivity. Memory determines dilution with the sixth power of the spatial scale.

Regular Transition from Contraction to Expansion

$$a(t) = a_b \left(1 + \frac{(t - t_b)^2}{\tau^2} \right)^{1/3}$$

$$a_b^3 = s_0/\rho_0, \quad \tau^2 = s_0/(6\pi G \rho_0^2)$$

For spatially flat dust, a vanishing effective cosmological constant, and $c = 1$, the scale reaches a positive minimum and the solution is regular at all times. The document proves causal completeness in this domain and separately studies local persistence under perturbations.

Constants, Fundamental Relations, and Physical Data

Coupling, Symmetries, and Closure of the Continuum

Numerical determination and exceptional incidence are readings of the same construction. Sources: monographs IV, XI, and XII.

Exceptional–Temporal Determination

$$19\,446 \longrightarrow 79 \longrightarrow 1$$

The selector traverses 19,446 candidates in the terminal domain; 79 satisfy exceptional compatibility, and one also satisfies the temporal condition. Hadamard inversion recovers the selected record in another representation: recovery and selection perform distinct functions within the same construction.

Conservation and Reversibility of Information

$$\|v\|^2 = \sum_{n \geq 0} \|m_n\|^2, \quad v = \sum_{n \geq 0} R_n^* D_n^* m_n$$

The bilateral law and the infinite archive preserve the norm and allow the state to be reconstructed. Phase return does not erase memory: restitution depends on preserving the relations of the complete sequence.

Marked Incidence, Leech, Moonshine, and M-Theory

$$[12, 6, 6]_3 \longrightarrow A_2^{12} \longrightarrow \Lambda_{24} \longrightarrow V^{\natural}$$

$$\mathrm{Tr}_{V^{\natural}} q^{L_0 - 1} = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \dots$$

The Witt flag organizes the gluing, and the marked vector of squared norm 54 determines the rootless Leech neighbour. The vertex operator construction leads to the Moonshine module, with the Monster as its automorphism group. The constant's record also determines a nonzero direction: its complement fixes the reduction $11 \rightarrow 10$ after the uniform mode is removed in $12 \rightarrow 11$. This polarization is coordinated with the $26 \rightarrow 24$ screen and T-duality. The prolongation towards M-theory transports the action and the identity $Q^2 = H$ through the stated isometries and domains; the physical identification preserves the specific conditions of that realization.

Real Line and Internal Cardinal Classification

$$\mathbb{R}_{\mathfrak{G}} \longleftrightarrow \mathcal{P}_{\mathfrak{G}}(\mathbb{N}), \quad |\mathbb{R}_{\mathfrak{G}}| = \aleph_1^{\mathfrak{G}}$$

Compatible cylinders construct the line without introducing it as an initial datum. Genealogical closure prolongs the record through subsets definable with parameters already produced. Condensation assigns ordinal codes to the internal reals; the coding of countable well-orders and their ternary lifting construct the inverse injection. Cantor–Bernstein establishes cardinal equality. The classification covers the internal subsets of this line: infinite subsets are countable or equinumerous with the constructed continuum. The domain preserves the memory of its construction and the two projections from which it arises.

Constants, Fundamental Relations, and Physical Data

Arithmetic Invariants · Construction and Meaning

The kernel extends through distinct readers of the same structure. Residues and torsion determine characters; their moments produce spectral constants. Regularization preserves the finite parts of those moments. Compatible cylinders then allow resolutions and information rates to be compared. The recognized formulas come at the end of these constructions. Sources: the integral corpus, chapters 41–45, and the systematic treatise.

Constant	Construction and Function of the Reader	Value or Evaluated Level
Apéry	Third moment of the spectral counter; the trace realizes $\zeta(3) = 4\pi^2 \log A_2$.	1.202056903159594285...
Euler–Mascheroni	Finite part of the regional decomposition $C_0 + C_1 + C_2$, after separating the divergent term.	0.577215664901532860...
Stieltjes	Hierarchy of finite parts of the regional reader; they are recognized as Laurent coefficients of ζ at 1.	Orders 1–6 in the corpus
Euler–Kronecker	The oriented tritic character χ_{-3} supplies the logarithmic correction $\gamma + L'(1,\chi)/L(1,\chi)$.	0.945497280871680703...
Glaisher–Kinkelin	First level of regularized spectral products: $\exp[1/12 - \zeta'(-1)]$.	1.282427129100622636...
Bendersky A_2	Second level of the same hierarchy: $\log A_2 = -\zeta'(-2) = \zeta(3)/(4\pi^2)$.	1.0309167521973921...
Meissel–Mertens	The prime-number reading separates repeated powers: $\gamma - \sum_{k \geq 2} P(k)/k$, where P is the prime zeta function.	0.2614972128476428...
Twin-Prime Product	Product of local exclusion factors for pairs differing by two: $p(p - 2)/(p - 1)^2$, $p > 2$.	0.6601618158468696...
Khinchin	The cylinders project onto continued fractions; their Gauss reading yields the typical geometric mean of the partial quotients.	2.685452001065306445...
Lévy	In the same Gauss reading, the typical rate of denominator growth: $\pi^2/(12 \log 2)$.	1.18656911041562545...
Lochs	Ratio between base-b resolution and Gauss entropy: $6 \log 2 \log b / \pi^2$.	Base 10: 0.970270114392...
Feigenbaum · Parameter	Dodecaphasic profile: uniform selection of superstable centres; δ_n converges to δF . Level-14 approximant:	4.669201610074725910...
Feigenbaum · Scale	In the same profile, ratio between successive spatial distances, level 14.	2.502907881009697542...

Khinchin, Lévy and Lochs describe rates for almost every trajectory under the Gauss measure. Parametric Feigenbaum convergence is established through the first-root selector and the renormalisation and transversality conditions of the time crystal. Level-14 values remain approximants; spatial scaling is a distinct reading.

Constants, Fundamental Relations, and Physical Data

Metrological Comparison in Common Units

Each row compares the same quantity in the same units. Δ is evaluation minus reference; Δ/u expresses the difference in units of the stated uncertainty. Exact SI references have zero uncertainty, so that quotient is not defined.

Quantity	HMT Evaluation	Reference	Δ	Δ/u
$1/\alpha \cdot 2018$	137.035999084000000000...	137.035999084(21) CODATA 2018	$+1.06741 \times 10^{-32}$	$+5.08289 \times 10^{-25}$
$1/\alpha \cdot 2022$	137.035999084000000000...	137.035999177(21) CODATA 2022	-9.30000×10^{-8}	-4.42857
Electron · 2022 MeV/c ²	0.510998950690000809...	0.51099895069(16) CODATA 2022	$+8.08608 \times 10^{-16}$	$+5.05380 \times 10^{-6}$
Electron · 2018 MeV/c ²	0.510998950690000809...	0.51099895000(15) CODATA 2018	$+6.90001 \times 10^{-10}$	+4.60001

The reciprocal $1/\alpha$ is compared directly with the same CODATA quantity. Against 2018, the relative difference is 7.78924×10^{-35} ; the experimental uncertainty of $1/\alpha$ is 2.1×10^{-8} . The tabulated decimal $\alpha = 0.0072973525693(11)$ is rounded: its absolute difference is 1.61990×10^{-14} and is not a separate measurement. The electron uses $uE = 1$ MeV. Sources: I, VI, and NIST/CODATA 2018 and 2022.

Reduced Action, Length, and Gravitation

The constructed values and references are expressed in common units. Δ is their numerical difference; each row preserves the action section used in the evaluation.

$$\begin{aligned} \hbar_{\text{geom}} &= \sqrt{H_5 \eta_{\text{ret}}} \, 10^{-34} \text{ J s}, & \hbar_{\text{SI}} &= \frac{\hbar_{\text{SI}}}{2\pi} \\ \hbar_{\text{ret}} &= \eta_{\text{ret}} \, 10^{-34} \text{ J s}, & L_\alpha &= \frac{5759}{23040} \alpha_{\text{HMT}}^{16} L^*, & G_{\text{ret}} &= \frac{c^3 L_\alpha^2}{\hbar_{\text{ret}}} \end{aligned}$$

Quantity and Unit	Constructed Value	Reference	Δ	Δ/u
Reduced Planck $\hbar_{\text{geom}}$ J s · geometric mean	$1.054571817280596765... \times 10^{-34}$	$1.054571817646156391... \times 10^{-34}$ Exact SI	-3.65560×10^{-44}	Not applicable
Planck $h \cdot \text{J s}$ Geometric mean	$6.626070147703121129... \times 10^{-34}$	$6.62607015 \times 10^{-34}$ Exact SI	-2.29688×10^{-43}	Not applicable
Returned action $\hbar_{\text{ret}}$ J s	$1.054571816915039061... \times 10^{-34}$	$1.054571817646156391... \times 10^{-34}$ Exact SI	$-7.311173301 \times 10^{-44}$	Not applicable
Marked length L_α m	$1.616254982625126330... \times 10^{-35}$	$1.616255(18) \times 10^{-35}$ CODATA 2022 IP	$-1.737487367 \times 10^{-43}$	-0.000965270759
Gravitation G_{ret} $\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ Marked length and returned action	$6.674299659414022706... \times 10^{-11}$	$6.67430(15) \times 10^{-11}$ CODATA 2022	$-3.405859773 \times 10^{-18}$	-0.00227057318

$\hbar_{\text{geom}}$ expresses the common scale; $\hbar_{\text{ret}}$ is the returned action used in G_{ret} . The exact reduced reference is $\hbar_{\text{SI}}/(2\pi)$. Rounding $\hbar_{\text{geom}}$ to nine significant figures gives $6.62607015 \times 10^{-34}$; the relative difference before rounding is -3.46643×10^{-10} . The gravitational evaluation uses $L_\alpha = (5759/23040) \alpha^{16} \times 1$ m, $\hbar_{\text{ret}} = \eta_{\text{ret}} \times 10^{-34}$ J s, and $c = 299792458$ m/s. Length and G agree with the references within their uncertainties: $u(\text{IP}) = 1.8 \times 10^{-40}$ m and $u(G) = 1.5 \times 10^{-15}$ $\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. CODATA derives its Planck length from G , so those references are related. Sources: II, the joint length–gravitation development, integral corpus equation (36.14), NIST/CODATA 2022, and BIPM.

Constants, Fundamental Relations, and Physical Data

Boltzmann · Spectral Construction and Thermal Reading

The coefficient is obtained as a relative intensity of the marked spectral reference. The thermal reading combines information per reference and conditioned energy; their derivative fixes temperature in the transported bases.

Quantity	Evaluation in the Realization	Reference or Relation
Boltzmann constant · B*(uΘ)/uΘ	$1.380649 \times 10^{-23} \text{ uE/u}\Theta$	$1.380649 \times 10^{-23} \text{ J/K} \cdot \text{SI}$
Thermal transducer	$B^*(u\Theta) = b^* \text{ uE}$	$\Theta = 1/(b^*\beta)$
Energy per information	$b^*\Theta_{\text{cycle,a}} = \hbar a \text{ c}/(L\alpha \log 3)$	$G_a b^* \Theta_{\text{cycle,a}} \log 3 = c^4 L \alpha$

The comparison uses $\text{uE} = 1 \text{ J}$ and $\text{u}\Theta = 1 \text{ K}$; both columns therefore express energy per unit temperature. The numerical coefficient b^* comes from the spectral realization described above. Ranks 1, 380, and 649, degrees 23 + 3n, and the rank-one reference specify the realization. Within it, additivity and conditioning determine the transducer and the free-energy minimum. The number exactly matches the SI numeral of Boltzmann’s constant; the physical identification retains these operations and the energy and temperature bases. Source: II, marked thermal composition and reading instrument; the SI reference is used after evaluation.

CKM Mixing and the Barbero–Immirzi Coefficient

The five calculated CKM coordinates lie within the PDG 2025 reference intervals. Sines, CP phase, and Jarlskog invariant are compared in the same parametrization. Barbero–Immirzi is shown with the theoretical Ghosh–Mitra reference. Δ denotes evaluation minus reference.

$$\begin{aligned} \theta_{12} &= 2A - 5h_\theta + 28\Delta_{\text{ang}}, & \theta_{23} &= 7h_\theta - \frac{25}{2}\Delta_{\text{ang}} \\ \theta_{13} &= A - 20h_\theta - \frac{2}{3}\Delta_{\text{ang}}, & \delta_{\text{CP}} &= 9A, & h_\theta &= C^*/6 \\ \gamma &= \frac{\pi AC^*}{180} + 12\Delta S_{90} - \Delta S_{120} \end{aligned}$$

Observable	HMT Evaluation	Reference	Δ	Δ/u
sin θ ₁₂	0.225007404757	0.22501 ± 0.00068 PDG 2025	−2.595243 × 10 ^{−6}	−0.00381653
sin θ ₂₃	0.041841757010	0.04183 (+0.00079; −0.00069)	+1.175701 × 10 ^{−5}	+0.0148823
sin θ ₁₃	0.003734601164	0.003732 (+0.000090; −0.000085)	+2.601164 × 10 ^{−6}	+0.0289018
δ _{CP} · rad	1.146265461116	1.147 ± 0.026 PDG 2025	−7.345389 × 10 ^{−4}	−0.0282515
J _{CKM}	3.1189723326633 × 10 ^{−5}	3.12 × 10 ^{−5} (+0.13; −0.12) × 10 ^{−5}	−1.027667 × 10 ^{−8}	−0.00856389
γ Barbero–Immirzi	0.274007672694459...	0.274066858232830... Ghosh–Mitra, state-counting equation	−5.918554 × 10 ^{−5}	Not applicable

Sources: article II and the PDG 2025 CKM review, equations (12.27)–(12.28). Δ/u uses the published uncertainty on the side selected by the sign of Δ. The sines are matrix parameters; matrix elements also contain cosine factors. For γ, the relative difference from Ghosh–Mitra is −2.15953 × 10^{−4}; this reference follows from its state-counting equation.

Constants, Fundamental Relations, and Physical Data

CKM Angles Expressed in Degrees

To make the angular comparison explicit, the same coordinates as in the preceding table are expressed in degrees. The θ references are obtained by applying arcsin to the PDG 2025 sines; the CP phase is converted from radians. These are not additional measurements.

Angle	HMT Evaluation (°)	PDG Reference (°)	Difference (°)
θ_{12}	13.003314	13.003466 (+0.039990; -0.039983)	-0.000153
θ_{23}	2.398056	2.397382 (+0.045304; -0.039568)	+0.000674
θ_{13}	0.213977	0.213828 (+0.005157; -0.004870)	+0.000149
δ_{CP}	65.676173	65.718259 ± 1.489690	-0.042086

The figures are rounded only for presentation. Angular uncertainties are obtained by transforming the endpoints of the published intervals; degrees are not identified with sines. Source: PDG 2025, CKM review, equations (12.27)–(12.28).

Electroweak Angular Reading and the On-Shell Reference

W and Z share the counter-angle and differ by one angle unit. Their angular ratio retains $\exp(-x)$, while the complete signatures also contain Δ_4 , s_{120} , and ζ_{270} . Subtracting (94, -1, 0, -2, 4) and (95, -1, -11, 0, -4) preserves these terms and changes the ratio. Electroweak mixing is distinct from CKM mixing.

$$x = 1000\alpha\pi/180, \quad D_A = e^{-2x}, \quad s_{W, \text{ang}}^2 = 1 - D_A$$
$$\frac{m_W^{[3]}}{m_Z^{[3]}} = \exp(-x + 11\Delta_4 - 2s_{120} + 8\zeta_{270}), \quad s_{W, [3]}^2 = 1 - \left(\frac{m_W^{[3]}}{m_Z^{[3]}}\right)^2$$

HMT Reading	PDG 2025 Reference	Numerical Difference
Angular: 0.224870880752844...	0.22342 ± 0.00009 PDG 2025	+1.450881 × 10 ⁻³
Complete signature: 0.223012769521055...	0.22342 ± 0.00009 on-shell	-4.072305 × 10 ⁻⁴

The second row uses the complete-signature masses in the final table: $m_W = 80378.964909704547...$ and $m_Z = 91187.533444313408...$ MeV/c². On-shell defines $s_W^2 = 1 - M_W^2/M_Z^2$ for physical masses in that convention; the PDG fit includes radiative effects. The differences are 16.1209 and -4.52478 reference uncertainties. They are numerical distances; the physical comparison retains the signature realization and pole or resonance convention. The angular determinant does not replace the complete mass ratio. Sources: VI, the narrative W/Z section, and PDG 2025 §§10.2 and 10.4.5.

Constants, Fundamental Relations, and Physical Data

Electromagnetic Constants in SI Units

The HMT evaluation and the references are expressed in the same units. Both CODATA editions are retained to show the agreement with 2018 and the change in the 2022 adjustment. Δ means evaluation minus reference.

$$Z_0 = \frac{2h_{\text{SI}}\alpha_{\text{HMT}}}{e_{\text{SI}}^2}, \quad \mu_0 = \frac{Z_0}{c_{\text{SI}}}, \quad \epsilon_0 = \frac{1}{Z_0 c_{\text{SI}}}$$

Quantity and unit	HMT Evaluation	Reference	Δ	Δ/u
$\mu_0 \cdot \text{H/m}$ 2018	$1.256637062121047789\dots \times 10^{-6}$	$1.25663706212(19) \times 10^{-6}$	$+1.04779 \times 10^{-18}$	+0.00551468
$\mu_0 \cdot \text{H/m}$ 2022	$1.256637062121047789\dots \times 10^{-6}$	$1.25663706127(20) \times 10^{-6}$	$+8.51048 \times 10^{-16}$	+4.25524
$\epsilon_0 \cdot \text{F/m}$ 2018	$8.854187812793002329\dots \times 10^{-12}$	$8.8541878128(13) \times 10^{-12}$	-6.99767×10^{-24}	-0.00538282
$\epsilon_0 \cdot \text{F/m}$ 2022	$8.854187812793002329\dots \times 10^{-12}$	$8.8541878188(14) \times 10^{-12}$	-6.00700×10^{-21}	-4.29071
$Z_0 \cdot \Omega$ 2018	376.730313667167610257...	376.730313668(57)	-8.32390×10^{-10}	-0.0146033
$Z_0 \cdot \Omega$ 2022	376.730313667167610257...	376.730313412(59)	$+2.55168 \times 10^{-7}$	+4.32487
$c \cdot \text{m/s}$	299792458	299792458 Exact SI	0	Not applicable

The documented realization uses α_{HMT} and the exact defining values of the SI: $h_{\text{SI}} = 6.62607015 \times 10^{-34} \text{ J s}$, $e_{\text{SI}} = 1.602176634 \times 10^{-19} \text{ C}$, and $c_{\text{SI}} = 299792458 \text{ m/s}$. Here e_{SI} is the elementary charge. CODATA enters subsequently, as a reference; the three electromagnetic comparisons depend on the same evaluation of α . Equality of c expresses the adopted SI convention, not an independent prediction. The normalized coefficients of the constitutive derivation are not compared directly with dimensional quantities. Sources: integral corpus, p. 119, and NIST/CODATA 2018–2022. Parentheses indicate standard uncertainty.

Constants, Fundamental Relations, and Physical Data

Mass Evaluations and Documentary References (MeV/c²)

The twenty entries are retained in the order of article VI. B denotes the electron beta coordinate; N, the null chart; F, the complete signature; A, the angular chart; Fc, the composite signature. Δm is the HMT – reference difference.

$$m_X^{[3]} = m_e^\beta \exp(n_A x + \frac{s_{cY}}{6} + k_{\Delta_4} \Delta_4 + b_{120} s_{120} + b_\zeta \zeta_{270}), \quad \zeta_{270} = 135 \Delta_4^2$$
$$m_X^{(0)} = e_0 \exp(n_A x + \frac{s_{cY}}{6}), \quad x = A_{\text{rad}}, \quad y = C_{\text{rad}}^*$$

State	HMT Value	Reference ± u	Δm	Reading
e±	0.510998950690000809	0.51099895069 ± 0.00000000016	0.000000000000000809	B · beta
μ±	105.658360294435829303	105.6583755 ± 0.0000023	-0.000015205564170697	F · signature
τ±	1776.860917631432295595	1776.93 ± 0.09	-0.069082368567704405	F · signature
u, anti-u	2.165924536173907663	2.16 ± 0.07	0.005924536173907663	A · angular
d, anti-d	4.679550162948874172	4.70 ± 0.07	-0.020449837051125828	A · angular
c, anti-c	1270.500838641911135286	1272.9 ± 4.5	-2.399161358088864714	A · angular
s, anti-s	92.940093907845640641	92.9 ± 0.7	0.040093907845640641	A · angular
t, anti-t	172597.479544019136261367	172600 ± 270	-2.520455980863738633	A · angular
b, anti-b	4190.119763299790218075	4186 ± 6	4.119763299790218075	A · angular
γ	0	< 10 ⁻²⁴ (bound)	—	N · null
g	0	No reference	—	N · null
W±	80378.964909704547024865	80362.5 ± 7.7	16.464909704547024865	F · signature
Z°	91187.533444313407844855	91187.9 ± 2.0	-0.366555686592155145	F · signature
H	125292.466042536133000433	125130 ± 110	162.466042536133000433	A · angular
π°	134.976724327872125739	134.976827767685 (s/u)	-0.000103439812874261	Fc · composite
π±	139.570485171572191375	139.570390983681 (s/u)	0.000094187891191375	Fc · composite
K±	493.677085649530612476	493.676599458041 (s/u)	0.000486191489612476	Fc · composite
K°	497.611186497750457359	497.610767531258 (s/u)	0.000418966492457359	Fc · composite
p, anti-p	938.271751546984898299	938.27208943 (s/u)	-0.000337883015101701	Fc · composite
n, anti-n	939.565810472507023313	939.56542194 (s/u)	0.000388532507023313	Fc · composite

Source: VI, “Realization of the Observables and Metrological Comparison”, table of twenty evaluations. The first equation corresponds to F/Fc; the second to A, with a basal normalization distinct from the electron beta normalization. The seventeen F/A/Fc readings retain their descriptors. s/u: uncertainty not supplied in the supplement, not zero uncertainty. Quarks require a scheme and scale; W, Z, and H require a resonance convention. The differences for τ, Z, and H use the rounded references in this table. No differences are calculated against a bound or an absent reference.